THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 6: **Triangles** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** similar	Same shape, possibly different size
2	**(a)** 6 cm	BPT: $\frac{2}{3} = \frac{4}{EC} \Rightarrow EC = 6$
3	**(c)** 80°	$\angle R = \angle C = 180 - 100 = 80^\circ$
4	**(b)** Incorrect — sides are proportional but angles may differ	Similar polygons need equal angles too
5	**(c)** AA	Two angles equal ⇒ AA
6	**(c)** 8 cm	$\frac{AB}{DE} = \frac{BC}{EF} \Rightarrow EF = 8$
7	**(a)** Student A is correct	$\frac{6}{3} = \frac{8}{4} = \frac{10}{5} = 2$
8	**(a)** 60 m	$\frac{12}{8} = \frac{h}{40}$
9	**(c)** $\frac{3}{8}$	$\frac{AE}{EC} = \frac{3}{5} \Rightarrow \frac{AE}{AC} = \frac{3}{8}$
10	**(b)** Basic Proportionality Theorem	$\frac{AD}{DB} = 1 \Rightarrow \frac{AE}{EC} = 1$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{PE}{EQ} = \frac{4}{4.5} = \frac{8}{9}$, $\frac{PF}{FR} = \frac{8}{9}$ — equal ⇒ $EF \parallel QR$ (converse of BPT)	1+1
12	(i) Any two squares; any two equilateral triangles (ii) A square and a rectangle; a square and a rhombus (non-square)	1+1
13	$\angle DOC = 180^\circ - 125^\circ = 55^\circ$; $\angle DCO = 180^\circ - 70^\circ - 55^\circ = 55^\circ$; $\angle OAB = \angle OCD = 55^\circ$	1+1
14	False. They are **similar** (SSS), not necessarily congruent; they are congruent only when the ratio is 1.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{x}{x-2} = \frac{x+2}{x-1} \Rightarrow x^2 - x = x^2 - 4 \Rightarrow x = 4$	1+1+1
16	In $\triangle AOB$ and $\triangle COD$: $\angle AOB = \angle COD$ (v.o.a.), $\angle OAB = \angle OCD$ (alternate) ⇒ $\triangle AOB \sim \triangle COD$ (AA) ⇒ $\frac{AO}{CO} = \frac{BO}{DO} \Rightarrow \frac{AO}{BO} = \frac{CO}{DO}$	1+1+1
17	$\angle R$ is common and $\angle RPQ = \angle RTS$ (given) ⇒ $\triangle RPQ \sim \triangle RTS$ (AA)	1+1+1
	OR: $\frac{AD}{AB} = \frac{AE}{AC}$ (since $AD = AE$, $AB = AC$) and $\angle A$ is common ⇒ $\triangle ADE \sim \triangle ABC$ (SAS)	1+1+1
18	Similar triangles (AA): $\frac{1.5}{1.6} = \frac{H}{2.4+1.6} \Rightarrow H = \frac{1.5 \times 4}{1.6} = 3.75$ m	1+1+1
	OR: $\triangle ADE \sim \triangle ABC \Rightarrow \frac{DE}{15} = \frac{AE}{9} = \frac{4}{12} = \frac{1}{3} \Rightarrow DE = 5$ cm, $AE = 3$ cm	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Given, To prove, figure (1). Join BE, CD; draw $EL \perp AB$, $DM \perp AC$. $\frac{ar(ADE)}{ar(BDE)} = \frac{AD}{DB}$, $\frac{ar(ADE)}{ar(DEC)} = \frac{AE}{EC}$ (2). $ar(BDE) = ar(DEC)$ (same base DE, between same parallels) ⇒ $\frac{AD}{DB} = \frac{AE}{EC}$ (2)	1+2+2
	OR: (i) $\angle A$ common, $\angle B = \angle M = 90^\circ$ ⇒ AA (2) (ii) Corresponding sides proportional (1) (iii) $\frac{AB}{AM} = \frac{BC}{MP} \Rightarrow MP = \frac{4 \times 9}{6} = 6$ cm (2)	2+1+2
20	(i) AA (both triangles right-angled, equal angles at mirror) (ii) $\frac{1.2}{1.5} = \frac{H}{6} \Rightarrow H = 4.8$ m (iii) $\frac{1.2}{x} = \frac{6.4}{6} \Rightarrow x = 1.125$ m	1+2+2
	OR: $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AD}{PM}$; $BD = \frac{BC}{2}$, $QM = \frac{QR}{2} \Rightarrow \frac{AB}{PQ} = \frac{BD}{QM} = \frac{AD}{PM} \Rightarrow \triangle ABD \sim \triangle PQM$ (SSS) ⇒ $\angle B = \angle Q$. With $\frac{AB}{PQ} = \frac{BC}{QR} \Rightarrow \triangle ABC \sim \triangle PQR$ (SAS)	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME — SET B

Chapter 6: **Triangles** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** similar	All angles 90°, sides proportional
2	**(a)** 7.5 cm	$\frac{4}{6} = \frac{5}{EC}$
3	**(b)** 36 cm	Perimeters are in the ratio of sides: $24 \times \frac{3}{2} = 36$
4	**(b)** Incorrect — the included angles must also be equal (SAS)	SAS needs the included angle
5	**(c)** SSS	All three sides proportional
6	**(b)** 60°	$\angle X = \angle P = 60^\circ$
7	**(a)** Student A is correct	$\frac{3}{4} = \frac{4.5}{6} = 0.75 \Rightarrow$ parallel
8	**(a)** 27 m	$\frac{6}{4} = \frac{h}{18}$
9	**(b)** 10 cm	$\frac{DE}{BC} = \frac{2}{5}$
10	**(b)** the same shape, not necessarily the same size	Definition of similarity

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{6}{DB} = \frac{8}{4} \Rightarrow DB = 3$ cm	1+1
12	If two angles of one triangle equal two angles of another, they are similar. Yes: 90° and 35° equal in both $\Rightarrow$ AA	1+1
13	$\frac{AE}{EC} = \frac{3}{4} \Rightarrow AE = \frac{3}{7} \times 15 = \frac{45}{7}$ cm ≈ 6.43 cm	1+1
14	$\angle C = 70^\circ$, $\angle Q = 60^\circ$. So $\angle A = \angle P$, $\angle B = \angle Q$, $\angle C = \angle R \Rightarrow \triangle ABC \sim \triangle PQR$ (AA). Claim is false.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$EQ = 1.10$, $FR = 2.20$; $\frac{PE}{EQ} = \frac{0.18}{1.10} = \frac{9}{55}$, $\frac{PF}{FR} = \frac{0.36}{2.20} = \frac{9}{55} \Rightarrow$ equal $\Rightarrow EF \parallel QR$ (converse of BPT)	1+1+1
16	Join AC meeting EF at G. In $\triangle ADC$, $EG \parallel DC \Rightarrow \frac{AE}{ED} = \frac{AG}{GC}$. In $\triangle CAB$, $GF \parallel AB \Rightarrow \frac{AG}{GC} = \frac{BF}{FC}$. Hence $\frac{AE}{ED} = \frac{BF}{FC}$.	1+1+1
17	In $\triangle ADC$ and $\triangle BAC$: $\angle ADC = \angle BAC$, $\angle C$ common $\Rightarrow$ similar (AA) $\Rightarrow \frac{CA}{CB} = \frac{CD}{CA} \Rightarrow CA^2 = CB \cdot CD$	1+1+1
	OR: $\angle ADB = \angle CEB = 90^\circ$ and $\angle B$ common $\Rightarrow \triangle ABD \sim \triangle CBE$ (AA)	1+1+1
18	$\frac{1.8}{1.2} = \frac{h}{24} \Rightarrow h = 36$ m; shadow of pole $= \frac{9 \times 1.2}{1.8} = 6$ m	2+1
	OR: $\frac{x+1}{3x-1} = \frac{x+3}{3x+4} \Rightarrow 3x^2 + 7x + 4 = 3x^2 + 8x - 3 \Rightarrow x = 7$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Given $\frac{AD}{DB} = \frac{AE}{EC}$; suppose $DE \not\parallel BC$, draw $DE' \parallel BC$. By BPT $\frac{AD}{DB} = \frac{AE'}{E'C} \Rightarrow \frac{AE}{EC} = \frac{AE'}{E'C} \Rightarrow \frac{AC}{EC} = \frac{AC}{E'C} \Rightarrow EC = E'C \Rightarrow$ E and E′ coincide $\Rightarrow DE \parallel BC$	1+2+2
	OR: (i) $\angle A = \angle C$ (opposite angles), $\angle AEB = \angle CBF$ (alternate, $AE \parallel BC$) $\Rightarrow$ AA (3) (ii) $\frac{AB}{CF} = \frac{AE}{CB} \Rightarrow CF = \frac{8 \times 9}{12} = 6$ cm (2)	3+2
20	(i) AA (ii) distance $= 6$ m: $\frac{4.8}{1.6} = \frac{6+x}{x} \Rightarrow x = 3$ m (iii) $\frac{4.8}{1.6} = \frac{d+4.5}{4.5} \Rightarrow d = 9$ m $\Rightarrow$ 6 s	1+2+2
	OR: $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{2BD}{2QM} = \frac{BD}{QM}$ and $\angle B = \angle Q \Rightarrow \triangle ABD \sim \triangle PQM$ (SAS) $\Rightarrow \frac{AB}{PQ} = \frac{AD}{PM}$	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.